

Task 1: Forecasting Across Heterogeneous Sensors

AI651 – Deep Learning for Space, Time and Graphs

Assignment 1

Student: _____

Student ID: _____

Fall 2026

Repository URL: _____

Abstract

This report evaluates context-dependent forecasting for a synthetic four-station production line. The model receives a 96-sample local history and predicts 48 future samples. Results compare shared, sensor-specific, and period-routed ridge regression with Raw Attention, moving-average decomposition, and a delay mixer. On the full-preset, single-seed run, period-routed ridge has the lowest validation error for both established and held-out populations, while the combined Autoformer-inspired model is the strongest neural model in aggregate. The report distinguishes validation-based choices from final test evaluation and documents the limitations of this one synthetic run.

Contents

1	Task, data, and evaluation protocol	3
2	Part 1: context-dependent forecasting	3
2.1	Response 1A: hypothesis, results, and interpretation	3
2.1.1	Output 1.1: validation comparison	3
2.2	Response 1B: why forecasting must depend on context	4
2.2.1	Outputs 1.2 and 1.3: context and Attention	4
3	Part 2: separating slow structure from repetition	5
3.1	Response 2: decomposition, width selection, and evidence	5
3.2	Output 2.3: decomposition ablation	6
4	Part 3: mixing by temporal delay	7
4.1	Response 3: delay aggregation and recurrence evidence	7
4.2	Output 3.2: recurrence diagnostic	8
5	Part 4: compare designs and deploy	8
5.1	Response 4(a): hypothesis	8
5.2	Response 4(b): comparison and horizon behavior	8
5.2.1	Output 4.1: 2-by-2 comparison	8
5.2.2	Output 4.2: horizon behavior	10
5.3	Response 4(c): deployment choice and test results	10
6	Conclusion	11
	Reproducibility and AI-use disclosure	11
	References	11

1 Task, data, and evaluation protocol

The simulated production line has four vibration stations sampled at 20 Hz. Products A, B, and C have respective nominal period ranges of 14–18, 21–27, and 28–36 samples. The signal combines a slow baseline/drift, a repeating vibration, and noise. Stations 1–3 are established; station 4 is held out from model fitting. A forecaster sees only one station’s 96-sample context and predicts the next 48 samples. Windows do not cross operating-regime boundaries.

Training, validation, and final test targets lie in the disjoint intervals $[0, 9600)$, $[9600, 12480)$, and $[12480, 15360)$, respectively. Contexts are anchored at their last observed value and scaled using established-station training data only; predictions are restored to physical units for evaluation. The saved notebook setup records PA1_PRESET=full, data seed 0, model seed 0, and CPU device. Neural runs report 6,000 optimization steps.

Metrics. RMSE is reported in m/s^2 and MSE in $(\text{m/s}^2)^2$. Lower is better. The validation set selects models and deployment choices; the final test set is shown only after those choices are made.

2 Part 1: context-dependent forecasting

2.1 Response 1A: hypothesis, results, and interpretation

Timing note. The hypothesis below was entered into the notebook after the saved experiment outputs existed. It is therefore reported transparently as a retrospective hypothesis, not as a preregistered prediction.

The expected ordering was period-routed ridge, then Raw Attention, then approximately tied shared and sensor-specific ridge on established stations. The same ordering was expected on station 4, without sensor-specific ridge. The rationale was that period routing estimates pace from the observed context and selects a fitted map, whereas fixed ridge maps compromise across changing paces; Attention must learn context-dependent mixing.

2.1.1 Output 1.1: validation comparison

Table 1: Output 1.1: full-preset validation results. Product entries are RMSE.

Model	Est. RMSE	Est. MSE	Held-out RMSE	Held-out MSE	Parameters	Fit (s)
Shared ridge	0.767	0.588	0.876	0.767	4,608	0.002
Sensor-specific ridge	0.768	0.590	unavailable	unavailable	13,824	0.003
Period-routed ridge	0.166	0.028	0.173	0.030	59,904	0.006
Raw	0.431	0.186	0.612	0.374	368,368	760.032
Attention						

Table 2: Output 1.1 product-wise validation RMSE by population.

Population	Model	Product A	Product B	Product C	Units
Established	Shared ridge	0.805	0.854	0.621	m/s^2
Established	Sensor-specific ridge	0.804	0.852	0.630	m/s^2
Established	Period-routed ridge	0.181	0.164	0.152	m/s^2
Established	Raw Attention	0.388	0.482	0.418	m/s^2
Held-out	Shared ridge	0.909	0.982	0.713	m/s^2
Held-out	Period-routed ridge	0.184	0.182	0.150	m/s^2
Held-out	Raw Attention	0.647	0.580	0.607	m/s^2

On established stations, the measured ordering is period-routed ridge < Raw Attention < shared ridge $\approx$ sensor-specific ridge. On station 4 it is period-routed ridge < Raw Attention < shared ridge; sensor-specific ridge is unavailable because no station-4 matrix is fitted. Relative to shared ridge, period routing lowers aggregate RMSE by about 78% on established stations and 80% on station 4. Raw Attention lowers it by about 44% and 30%, respectively.

Across products, routed ridge and Raw Attention retain their relative ordering. Shared and sensor-specific ridge are near-tied and swap order: sensor-specific is marginally better for A and B, while shared is better for C. Station-specific fitting does not solve pace changes within a station. Period routing provides a useful explicit inductive bias by estimating period from the current context and choosing a pace-band map. Attention also depends on context, but in this run its flexibility does not beat period routing.

2.2 Response 1B: why forecasting must depend on context

2.2.1 Outputs 1.2 and 1.3: context and Attention

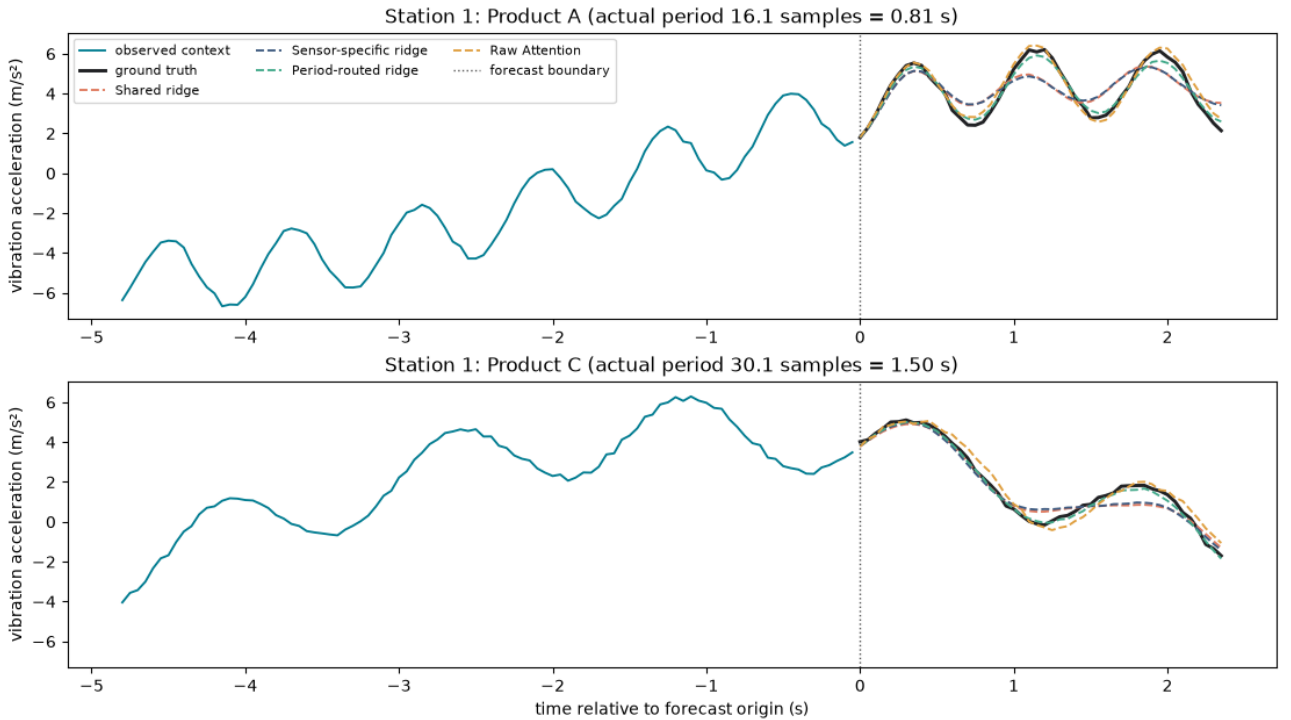


Figure 1: Output 1.2: forecasts for Station 1 under Product A and Product C, with different actual periods (16.135 and 30.093 samples). The fixed ridge curves compromise in phase/shape; the context-routed forecast follows the continuations more closely.

In the two plotted validation windows, shared-ridge window RMSE is 0.779 for Product A and 0.448 for Product C; period-routed ridge gives 0.232 and 0.150, and Raw Attention gives 0.290 and 0.378. These are individual examples, not aggregate performance estimates. The same station faces different periods, so station identity alone cannot specify the phase advance needed for the next 48 samples. A fixed linear map must compromise across these incompatible continuations.

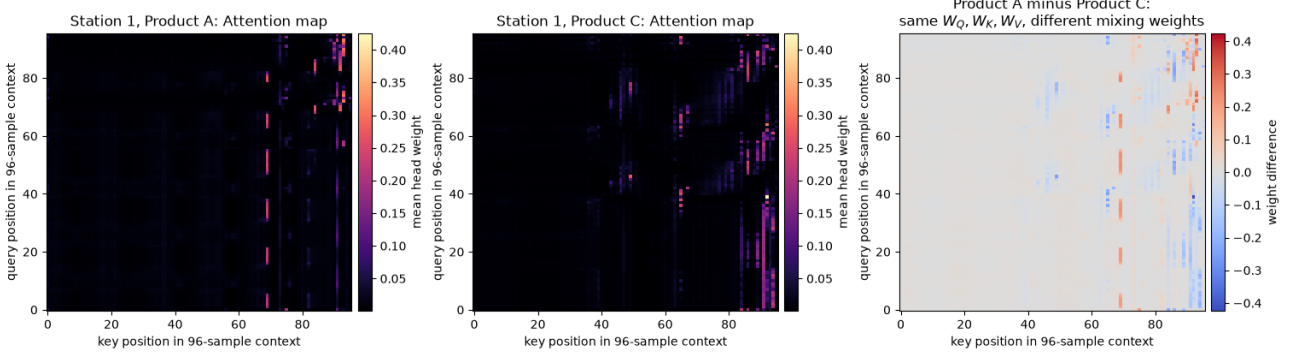


Figure 2: Output 1.3: learned Attention weights respond differently to Product A and Product C contexts. This demonstrates context dependence, not recovery of the physical period.

The learned query, key, and value projections are fixed after training, but their outputs depend on the input context. Thus the query-key scores and softmax weights, and therefore the value-mixing operation, change from one context to another. The saved mean absolute weight difference is 0.0113 and the maximum is 0.4250. This does not establish that Attention has recovered the true period; it shows only that its mixing weights depend on the input.

3 Part 2: separating slow structure from repetition

3.1 Response 2: decomposition, width selection, and evidence

For odd width $m = 2q + 1$, the trend is the centered moving average

$$T_t = \frac{1}{m} \sum_{u=-q}^q x_{t+u}, \quad S_t = x_t - T_t.$$

At boundaries the nearest endpoint is replicated. The streams have the same length and satisfy $S + T = x$. The trend is sent to a linear forecasting head, while the remainder is embedded and processed by the encoder; decomposition also follows each residual sublayer.

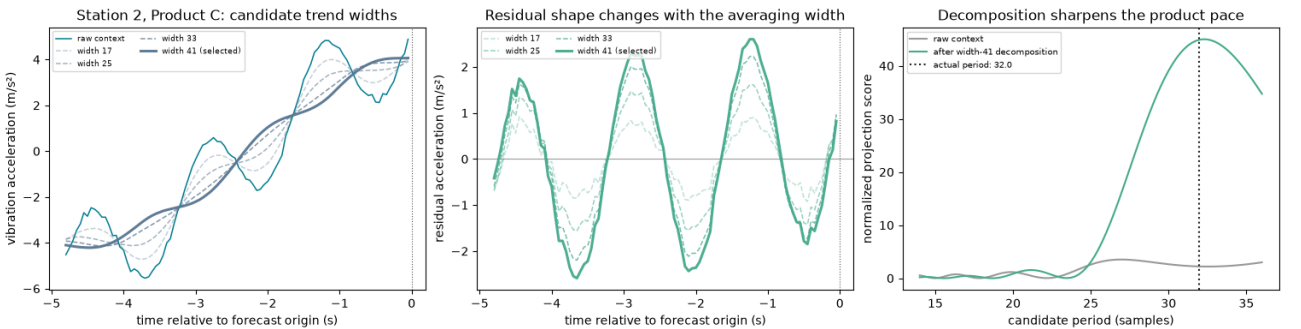
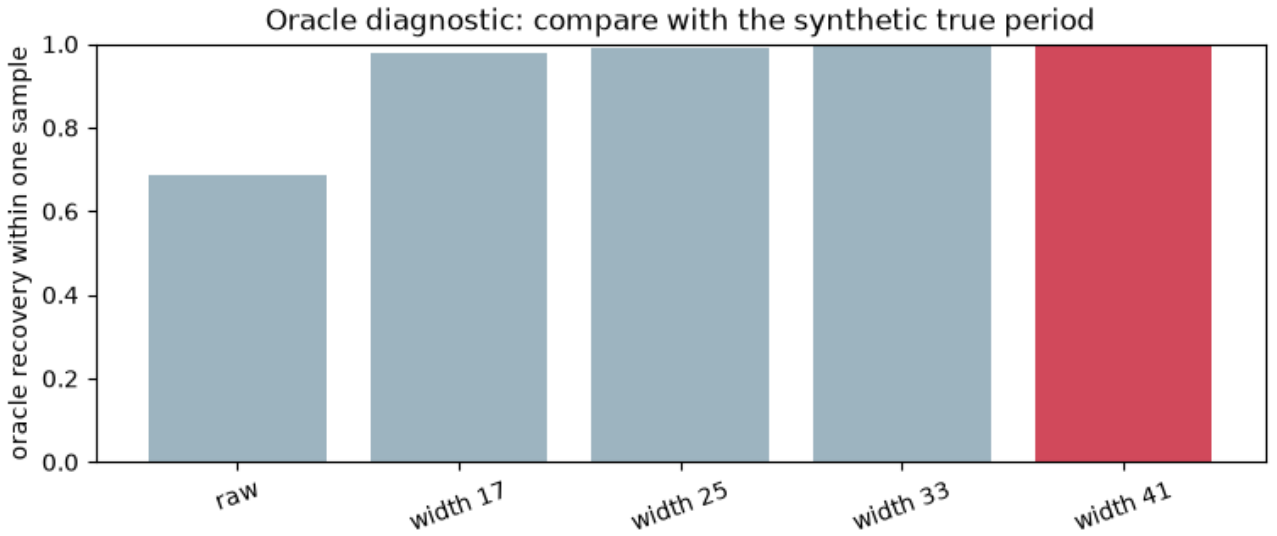


Figure 3: Output 2.1: candidate moving-average widths on a Station 2/Product C context. Wider width 41 leaves a stronger oscillatory residual and yields a clear period peak after decomposition.

Table 3: Output 2.1: candidate trend widths and training-only oracle diagnostic.

Width	Selected	Residual std (m/s ²)	Trend range (m/s ²)	Recovery within 1 sample
17	No	0.578	8.768	98.00%
25	No	1.012	8.037	98.92%
33	No	1.372	8.076	99.33%
41	Yes	1.572	8.280	99.85%
Raw	–	–	–	68.67%

Width 41 (2.05 seconds at 20 Hz) was selected by the highest training-only oracle recovery within one sample. It is not the minimum-median-error width: width 25’s median absolute period error is 0.122 samples versus 0.216 for width 41. The oracle diagnostic uses the known synthetic period only for training-window preprocessing selection/diagnosis; it is not fed to the forecaster and would not be available as such in ordinary deployment.

**Figure 4:** Output 2.2: oracle period recovery within one sample for the raw context and four decomposition widths. The selected width maximizes this criterion.

3.2 Output 2.3: decomposition ablation

Table 4: Output 2.3: validation errors by population and product.

Model	Population / product	MSE (m/s ²) ²	RMSE (m/s ²)	MAE (m/s ²)
Raw Attention	Established / A	0.150	0.388	0.293
Raw Attention	Established / B	0.233	0.482	0.358
Raw Attention	Established / C	0.175	0.418	0.306
Raw Attention	Held-out / A	0.418	0.647	0.502
Raw Attention	Held-out / B	0.336	0.580	0.439
Raw Attention	Held-out / C	0.369	0.607	0.456
Attention + decomposition	Established / A	0.070	0.264	0.204
Attention + decomposition	Established / B	0.112	0.335	0.249
Attention + decomposition	Established / C	0.080	0.283	0.212
Attention + decomposition	Held-out / A	0.129	0.359	0.275
Attention + decomposition	Held-out / B	0.238	0.488	0.357
Attention + decomposition	Held-out / C	0.189	0.434	0.330

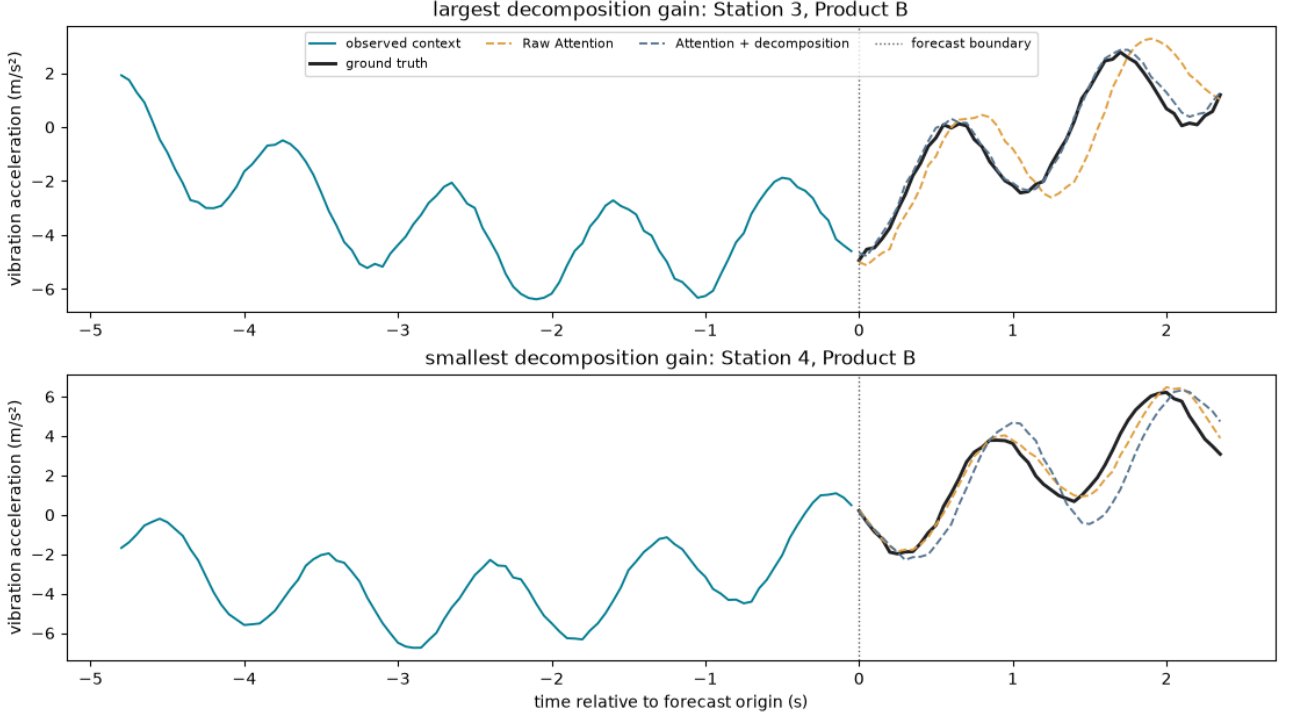


Figure 5: Output 2.3: validation ablation showing the effect of decomposition on forecasts and errors.

On aggregate validation, decomposition lowers Attention RMSE from 0.431 to 0.296 on established stations and from 0.612 to 0.430 on station 4, reductions of about 31% and 30%. It improves each product/population row in the ablation. A moving average is sensible for the synthetic world because the baseline varies smoothly on a longer time scale than the vibration cycle. An abrupt baseline jump would violate this assumption: averaging would smear the transition and leave transient structure in the residual, where it may be mistaken for repetition.

4 Part 3: mixing by temporal delay

4.1 Response 3: delay aggregation and recurrence evidence

For each selected delay τ_j and weight α_j , the aggregation implements

$$z_t = \sum_{j=1}^K \alpha_j v_{(t-\tau_j) \bmod L}.$$

The implementation vectorizes selection with circular indices and shares each example’s delays and weights across heads and features. The immediate worked example returns $[35, 15, 25, 25]$ for values $[10, 20, 30, 40]$, delays $[1, 3]$, and weights $[0.75, 0.25]$. This verifies the required direction: $z_0 = 0.75(40) + 0.25(20) = 35$.

Table 5: Output 3.1: supplied FFT delay scores equal direct circular correlation.

Delay	FFT score	Direct score
0	4	4
1	-4	-4
2	4	4
3	-4	-4

4.2 Output 3.2: recurrence diagnostic

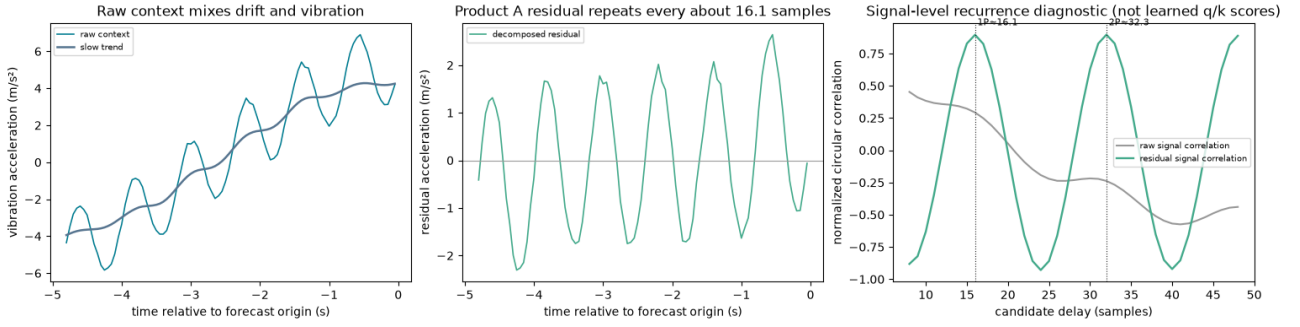


Figure 6: Output 3.2: raw and decomposed-signal circular recurrence for Station 2/Product A. The diagnostic is computed from signal values, not learned query/key scores.

For the plotted run, the actual period is 16.135 samples. Residual recurrence peaks at integer delays 16 and 32, near P and $2P$, with correlations 0.896 and 0.897; the next peak is near 48, approximately $3P$. The raw signal’s slow drift creates broad, less period-specific correlation. Removing the smooth component makes the repeating structure more visible and supports a mixer organized by displacement. This is a signal-level diagnostic, not evidence that the trained model’s learned query/key scores peak at those delays.

Delay mixing would be a poor assumption for an isolated transient: it has no recurring displacement, and circularly shifted copies can wrap or spread the event to unrelated positions.

5 Part 4: compare designs and deploy

5.1 Response 4(a): hypothesis

Timing note. As with Response 1A, this prediction was entered after saved experiment outputs existed and is presented as a retrospective hypothesis, not as a preregistered prediction. The hypothesis was that decomposition should help most when the slow component is prominent, because it removes the ramp from the encoder and can expose recurring structure. Delay mixing alone was expected to benefit less directly because it still scores projected features from a raw representation.

5.2 Response 4(b): comparison and horizon behavior

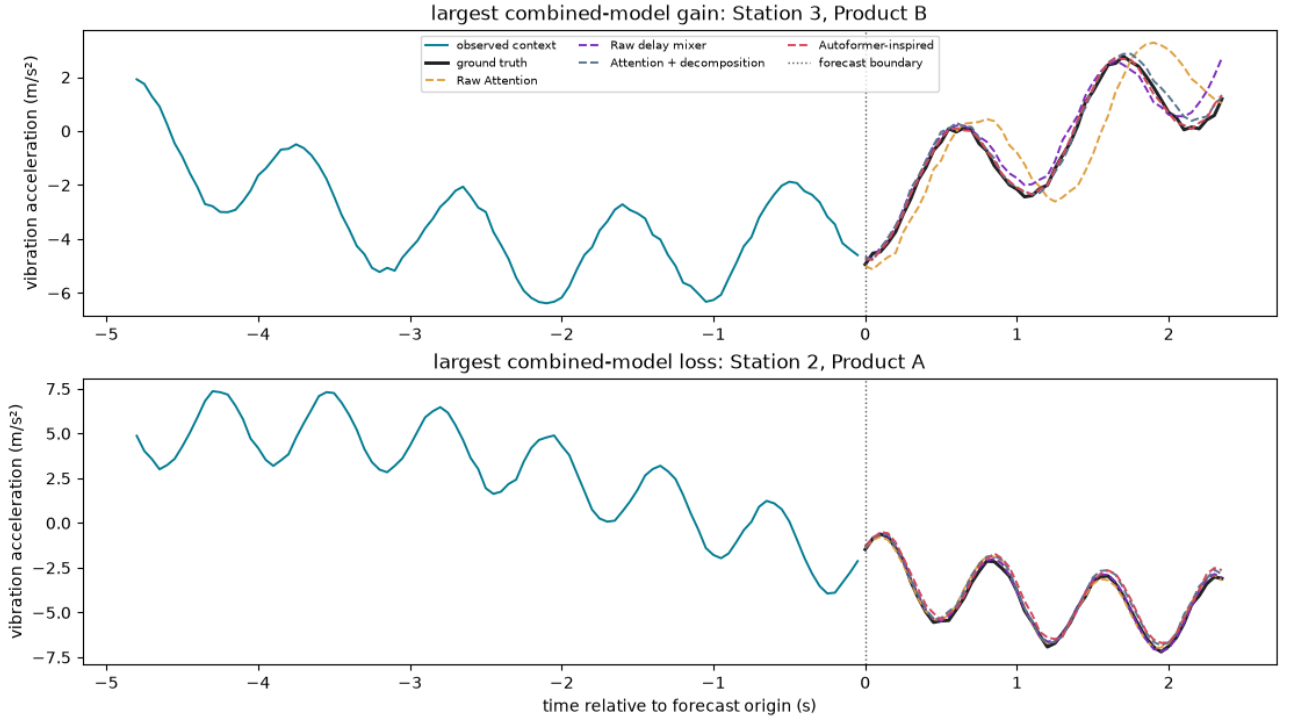
5.2.1 Output 4.1: 2-by-2 comparison

Table 6: Output 4.1: all four neural configurations on validation.

Model	Est. RMSE	Est. MSE	Held-out RMSE	Held-out MSE	Parameters	Fit (s)
Raw	0.431	0.186	0.612	0.374	368,368	760.032
Attention	0.296	0.087	0.430	0.185	373,024	897.249
Attention + decomposition	0.221	0.049	0.340	0.116	368,370	899.230
Raw delay mixer	0.199	0.040	0.239	0.057	373,026	1008.758
Autoformer-inspired						

Table 7: Output 4.1: product-wise validation RMSE for the four neural configurations.

Population	Model	A	B	C
Established	Raw Attention	0.388	0.482	0.418
Established	Attention + decomposition	0.264	0.335	0.283
Established	Raw delay mixer	0.211	0.211	0.239
Established	Autoformer-inspired	0.226	0.188	0.182
Held-out	Raw Attention	0.647	0.580	0.607
Held-out	Attention + decomposition	0.359	0.488	0.434
Held-out	Raw delay mixer	0.415	0.281	0.309
Held-out	Autoformer-inspired	0.270	0.221	0.224

**Figure 7:** Output 4.1: contrasting validation windows with the largest combined-model gain and loss. The combined model is not best on every individual window.

Both switches improve aggregate validation error. For established stations, decomposition lowers MSE by 0.099 with Attention and 0.009 with delay mixing; replacing Attention with delay mixing lowers it by 0.137 on raw inputs and 0.047 on decomposed inputs. For station 4, the corresponding reductions are 0.189/0.059 and 0.258/0.128. Delay mixing alone has the larger aggregate gain, but these scores do not isolate the drift-prominent contexts targeted by the hypothesis. Combined gains (0.146 established, 0.317 held-out) are smaller than the sums of individual gains (0.236/0.447). This is consistent with overlapping benefits in this run, not proof of a causal mechanism.

Autoformer-inspired is the strongest neural model in aggregate, but not on every product/window. Established Product A favors raw delay mixing (RMSE 0.211 vs. 0.226). In the largest-loss contrast, raw delay mixing has RMSE 0.139 versus 0.358 for Autoformer-inspired.

5.2.2 Output 4.2: horizon behavior

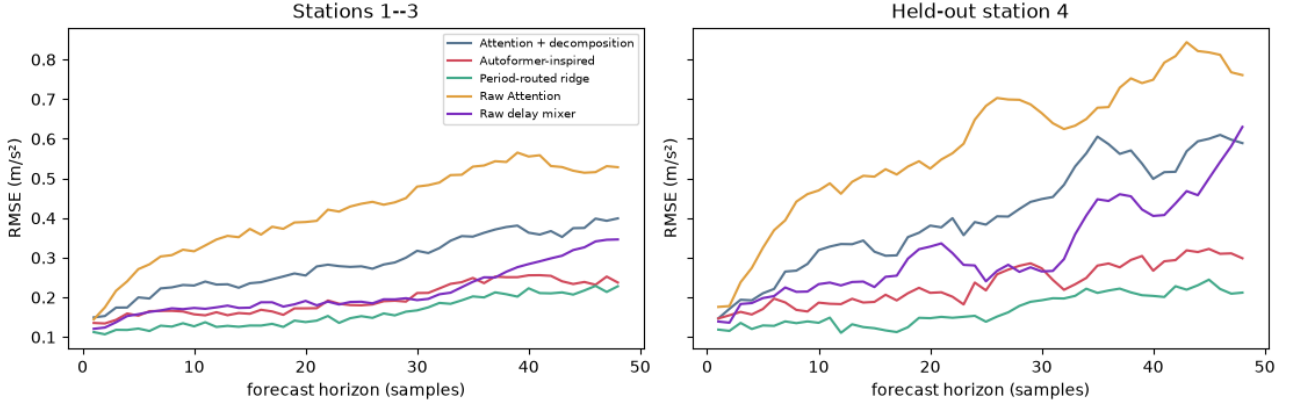


Figure 8: Output 4.2: RMSE across the 48-step forecast horizon for established and held-out populations.

Table 8: Output 4.2: RMSE at forecast steps 1 and 48 (validation, m/s^2).

Population	Model	Step 1	Step 48
Established	Period-routed ridge	0.112	0.228
Established	Raw Attention	0.144	0.528
Established	Raw delay mixer	0.121	0.346
Established	Attention + decomposition	0.149	0.399
Established	Autoformer-inspired	0.135	0.237
Held-out	Period-routed ridge	0.119	0.212
Held-out	Raw Attention	0.176	0.761
Held-out	Raw delay mixer	0.139	0.630
Held-out	Attention + decomposition	0.147	0.589
Held-out	Autoformer-inspired	0.147	0.299

Errors generally rise with forecast horizon, though not monotonically. The increase is steepest for Raw Attention and Raw delay mixer, particularly on the held-out station. Autoformer-inspired and period-routed ridge remain comparatively stable toward step 48.

5.3 Response 4(c): deployment choice and test results

Model choice was made using validation evidence. Period-routed ridge is selected for both populations because it has the lowest validation error, 59,904 parameters, and a recorded fitting time of approximately 0.006 seconds, far below the neural fits. Sensor-specific ridge cannot serve station 4 because it has no fitted model for that station.

Table 9: Output 4.3: selected model validation and final test results.

Population	Model	Val. MSE	Val. RMSE	Test MSE	Test RMSE
Established	Period-routed ridge	0.028	0.166	0.026	0.161
Held-out	Period-routed ridge	0.030	0.173	0.033	0.183

The close validation/test RMSEs support generalization within this simulated setting. Test scores are reported as the final evaluation of a validation-selected model, not used to select it.

6 Conclusion

The experiment illustrates the value of inductive bias. Period-routed ridge is substantially better than the other listed models on both validation populations despite being much smaller and faster to fit. Among neural variants, decomposition and delay mixing each improve aggregate validation accuracy relative to Raw Attention, and their combination performs best overall. However, the improvements are subadditive in this run and individual product/window exceptions remain. These observations support the methods on this synthetic benchmark; they do not establish universal superiority or real-world equipment-monitoring performance.

Reproducibility and AI-use disclosure

The figures in this report are the eight saved PNG outputs extracted from the executed full-preset notebook. Numerical tables reproduce its saved CSV outputs. No new training or notebook execution was performed to prepare this report. The recorded run uses data seed 0 and model seed 0; results are therefore a single-seed experiment.

Generative AI assistance (Copilot) was used to help inspect Task 1 requirements, compare notebook outputs with those requirements, extract notebook figures, and draft/organize explanatory report text. Representative prompts requested an implementation/results audit, comparison of hypotheses with validation results, and a comprehensive LaTeX report. The AI output included draft summaries, interpretations, tables, and report structure. The hypotheses were supplied by the student; the report explicitly identifies them as retrospective because they were added after saved outputs existed. The student revised the 1A ordering language and supplied/revised the 4A drift-focused hypothesis; the AI draft was adjusted to reflect those edits and the saved validation values. The AI did not run training or generate the experimental measurements. The student should review this report, verify all claims, and disclose any further edits or AI use in accordance with course policy.

References

1. Wu, H. et al. (2021). “Autoformer: Decomposition Transformers with Auto-Correlation for Long-Term Series Forecasting,” Sections 3.1–3.2.
2. Vaswani, A. et al. (2017). “Attention Is All You Need.”
3. Zeng, A. et al. (2023). “Are Transformers Effective for Time Series Forecasting?”
4. Hyndman, R. J. and Athanasopoulos, G. *Forecasting: Principles and Practice*, Section 5.10.
5. AI651 Assignment 1, “Forecasting Across Heterogeneous Sensors,” Fall 2026; supplied assignment notebook and harness.